

SAFWAN SABIR

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PROFILE

Full-stack engineer with 2+ years across software development, spanning AI/ML systems, cloud infrastructure, and production web platforms. Built and deployed multi-agent LLM pipelines, RAG architectures, and multi-tenant AWS infrastructure serving live workloads. Proficient in Python, TypeScript, and JavaScript, with hands-on experience in async backend systems, geospatial data processing, and CI/CD automation. Seeking to apply full-stack and AI engineering skills to production-scale systems.

EDUCATION

University of Central Punjab
Bachelor of Science in Computer Science

Lahore, Pakistan

EXPERIENCE

Devsinc
Full-time, On-site — 9 months total

Lahore, Pakistan

Associate Software Engineer

March 2026 – Present

- Containerize model inference services with Docker and deploy to AWS, maintaining reproducible environments across development, staging, and production.
- Build automated model evaluation scripts using Pandas, NumPy, and Matplotlib, generating precision, recall, and latency reports for 5+ deployed models.
- Integrate ML model outputs with REST APIs in collaboration with backend teams, supporting downstream data analysis and reporting workflows.

Trainee Associate Software Engineer

January 2026 – March 2026

- Built scalable ETL pipelines using PySpark and Databricks to process large-scale structured datasets, reducing data preparation time by approximately 30%.
- Fine-tuned transformer models on domain-specific data using PyTorch and Hugging Face, improving task-specific accuracy by over 12% versus baseline.

AI Intern

December 2025 – January 2026

- Developed and deployed a full-stack ML pipeline integrating GPT-based language models with a FastAPI backend, enabling real-time inference under production load.

UmrahXpress

London, United Kingdom (Remote)

Part-time — 1 year 3 months total

Full-Stack Developer

February 2025 – December 2025

- Architected an internal B2B hotel price-comparison platform aggregating live rates across 4 third-party supplier APIs (Hotelbeds, RateHawk, TBO Holidays, StubA) using a FastAPI backend with concurrent async fan-out requests.
- Designed a custom hotel deduplication engine using GPS-proximity matching and fuzzy text comparison on PostgreSQL with PostGIS, eliminating duplicate listings across 4 supplier feeds without a paid third-party mapping subscription.
- Implemented role-based access control across 3 permission tiers (Admin, Agent, Manager), separating search/quote access from booking-execution rights.
- Built a booking-data synchronization system combining supplier webhooks with a 12-hour reconciliation cron job to keep internal booking records consistent with live supplier statuses.

- Introduced Redis caching for volatile hotel rate data, cutting redundant calls to 4 external supplier APIs during repeated searches.
- Maintained the company's public-facing landing site (umrahxpress.co.uk) using React and Next.js.

Junior Full-Stack Developer

October 2024 – January 2025

- Contributed frontend features for the company's public-facing landing site (umrahxpress.co.uk) using React and Next.js.
- Assisted in early-stage backend development of the internal CRM system using Python and FastAPI.

Hashtag Marketing

Lahore, Pakistan (Hybrid)

Full-time — 1 year 2 months

Graphic Designer

February 2022 – March 2023

- Designed social media marketing creatives and promotional content for 15+ concurrent client accounts spanning food & beverage, retail, travel, and real estate sectors.
- Produced static and motion content using Adobe Photoshop, Illustrator, After Effects, and Premiere Pro, including product photography compositions, branded social posts, and short-form video edits.
- Maintained brand consistency and rapid turnaround times across a fast-paced agency production schedule.

PROJECTS

THOR AI Platform (Zenhub / Octopus) — Cloud Infrastructure Engineer

- Provisioned multi-tenant production AWS infrastructure using Terraform (remote state on S3 with DynamoDB locking) and Kubernetes via Amazon EKS v1.34, across 3 mixed node groups.
- Deployed per-tenant RDS PostgreSQL instances and ElastiCache Redis replication groups (1 primary + 2 replicas) with KMS encryption at rest and TLS in transit.
- Configured CloudFront as a reverse proxy backed by AWS WAF v2 (IP reputation, OWASP Core Rule Set, SQL-injection rules), with a separate regional WAF protecting the EKS ingress.
- Built a zero-static-credentials CI/CD pipeline using AWS CodeBuild triggered via GitHub Actions with OIDC-federated IAM roles for Docker builds and Kubernetes rollouts.
- Automated pull-request preview environments with ephemeral Kubernetes namespaces, RDS snapshot restores, and scoped IAM roles with ECR image lifecycle policies.

VITAL — Virtual Interactive Training for Aspiring Clinicians

- Architected a multi-agent AI clinical training system using LangGraph and FastAPI to orchestrate specialized ReAct sub-agents for zero-hallucination medical training with automated, evidence-based feedback.
- Built an immersive 3D WebGL interface with React, Vite, and Three.js, rendering interactive VRM avatars with real-time lip-sync for simulated patient encounters.
- Engineered a deterministic RAG pipeline using ChromaDB and PostgreSQL, anchoring LLM outputs to verifiable clinical metrics and mitigating hallucination risk.
- Directed a full backend migration from OpenAI to Google Gemini and containerized the microservices architecture with Docker Compose across Vercel and Koyeb.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Frameworks & Libraries: React, Next.js, FastAPI, LangGraph, PyTorch, Hugging Face, Three.js, Pandas, NumPy, Scikit-learn

AI & Machine Learning: LLM Integration, RAG, Multi-Agent Systems, Model Context Protocol (MCP), GPT/Gemini Integration, Fine-Tuning, Embeddings (ChromaDB)

Cloud & Infrastructure: AWS (EKS, RDS, WAF, Route 53, CloudFront, ACM, IAM/OIDC), Terraform, Kubernetes, Docker, Vercel, Koyeb, Cloudflare R2, Redis, PostgreSQL/PostGIS, Databricks

Tools & Practices: Git, GitHub Actions, CI/CD, RESTful APIs, WebSockets, Microservices, Clerk, Pydantic, Vitest